MATH 159 (Calculus II): Complete Definitions, Laws, Theorems & Short Questions Solutions

Section A: Complex Variables · Section B: Vector Analysis & Vector Calculus
With Full Proofs, Working Formulas & Textbook Citations
Based on BUET L-1/T-1 EEE Examinations (2010–2025)

Comprehensive BUET Exam Preparation Reference

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Part I

Section A: Complex Variables (Definitions, Theorems, Proofs & Short Questions)

1 Complex Numbers, Triangle Inequalities, Loci & Multi-Valued Functions

1.1 Triangle Inequalities & Modulus-Argument Short Proofs

Exam Appearances: 2010-11 Q1a,b,c, 2014-15 Q1c, 2016-17 Q1b, Q2b, 2017-18 Q1a, 2021-22 Q1b, 2022-23 Q1a, Q2c, 2023-24 Q1a.
Textbook Citations: J. W. Brown & R. V. Churchill, *Complex Variables and Applications* (9th Ed.), Ch. 1, Sec. 4–9 & Ch. 3, Sec. 30–35; M. R. Spiegel, *Schaum's Outline of Complex Variables* (2nd Ed.), Ch. 1.

1. Key Working Formulas for Complex Numbers

$$z = x + iy = r(\cos\theta + i\sin\theta) = re^{i\theta}, \qquad \bar{z} = x - iy = re^{-i\theta}, \qquad |z|^2 = z\bar{z} = x^2 + y^2$$

$$\operatorname{Re}(z) = \frac{z+\bar{z}}{2} \le |z|, \qquad \operatorname{Im}(z) = \frac{z-\bar{z}}{2i} \le |z|, \qquad z^{1/n} = r^{1/n}\exp\left[i\left(\frac{\theta_0 + 2k\pi}{n}\right)\right] \quad (k = 0, 1, \ldots, n-1)$$

$$\operatorname{Ln} z = \ln|z| + i\operatorname{Arg} z \quad (-\pi < \operatorname{Arg} z \le \pi), \qquad \ln z = \ln|z| + i(\operatorname{Arg} z + 2n\pi) \quad (n \in \mathbb{Z})$$

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i} = \sin x\cosh y + i\cos x\sinh y, \qquad \cos z = \frac{e^{iz} + e^{-iz}}{2} = \cos x\cosh y - i\sin x\sinh y$$

$$\sinh z = -i\sin(iz) = \sinh x\cos y + i\cosh x\sin y, \qquad \cosh z = \cos(iz) = \cosh x\cos y + i\sinh x\sin y$$

2. Prove the Triangle Inequalities: (i) $|z_1 + z_2| \le |z_1| + |z_2|$ and (ii) $|z_1 - z_2| \ge \big||z_1| - |z_2|\big|$ (2017-18 Q1a, 2023-24 Q1a)

Proof of (i): Using $|z|^2 = z\bar{z}$ and $w + \bar{w} = 2\operatorname{Re}(w) \le 2|w|$:

$$\begin{aligned} |z_1 + z_2|^2 &= (z_1 + z_2)(\bar{z}_1 + \bar{z}_2) = z_1\bar{z}_1 + (z_1\bar{z}_2 + \bar{z}_1 z_2) + z_2\bar{z}_2 \\ &= |z_1|^2 + 2\operatorname{Re}(z_1\bar{z}_2) + |z_2|^2 \\ &\le |z_1|^2 + 2|z_1\bar{z}_2| + |z_2|^2 = |z_1|^2 + 2|z_1||z_2| + |z_2|^2 = (|z_1| + |z_2|)^2 \end{aligned}$$

Taking the non-negative square root of both sides gives:

$$\boxed{|z_1 + z_2| \le |z_1| + |z_2|} \tag{1}$$

Proof of (ii): Write $z_1 = (z_1 - z_2) + z_2$ and apply part (i):

$$|z_1| = |(z_1 - z_2) + z_2| \le |z_1 - z_2| + |z_2| \implies |z_1 - z_2| \ge |z_1| - |z_2|$$

Interchanging z_1 and z_2 gives $|z_2 - z_1| = |z_1 - z_2| \ge |z_2| - |z_1| = -(|z_1| - |z_2|)$. Combining both inequalities yields:

$$\boxed{|z_1 - z_2| \ge \big||z_1| - |z_2|\big|} \tag{2}$$

3. Standard Short Proofs on Moduli, Arguments & Logarithms

- **Problem 1 (2010–11 Q1a):** *If $|z+ia| = |z+ib|$ ($a, b \in \mathbb{R}, a \neq b$), show that $z - \bar{z} = -i(a+b)$.*
  Solution: Squaring both sides and using $|w|^2 = w\bar{w}$ with $\overline{ia} = -ia$:

$$(z+ia)(\bar{z}-ia) = (z+ib)(\bar{z}-ib)$$
$$z\bar{z} - ia(z-\bar{z}) + a^2 = z\bar{z} - ib(z-\bar{z}) + b^2$$
$$-i(a-b)(z-\bar{z}) = b^2 - a^2 = -(a-b)(a+b)$$

  Dividing by $(a-b) \neq 0$ and multiplying by $-i$ (since $\frac{1}{i} = -i$) gives $\boxed{z - \bar{z} = -i(a+b)}$.

- **Problem 2 (2021–22 Q1b):** *If $|z^2 - 1| = |z^2| + 1$, show that z lies on the imaginary axis.*
  Solution: Squaring both sides gives $(z^2-1)(\bar{z}^2-1) = (|z|^2+1)^2$:

$$|z|^4 - (z^2 + \bar{z}^2) + 1 = |z|^4 + 2|z|^2 + 1 \implies z^2 + \bar{z}^2 + 2z\bar{z} = 0 \implies (z+\bar{z})^2 = 0$$

  Thus $z + \bar{z} = 2\,\mathrm{Re}(z) = 0 \implies x = \mathrm{Re}(z) = 0$, proving that $z = iy$ lies on the **imaginary axis**.

- **Problem 3 (2016–17 Q1b):** *Show that $|\,\mathrm{Re}(2 + \bar{z} + z^3)| \leq 4$ when $|z| \leq 1$.*
  Solution: Using $|\,\mathrm{Re}(w)| \leq |w|$ and the triangle inequality for $|z| \leq 1$:

$$|\,\mathrm{Re}(2+\bar{z}+z^3)| \leq |2+\bar{z}+z^3| \leq 2 + |\bar{z}| + |z|^3 \leq 2 + 1 + 1^3 = \boxed{4}$$

- **Problem 4 (2022–23 Q1a):** *Calculate $\mathrm{Ln}\, z$ for $z = i$ and show $|\sin z|^2 = \sin^2 x + \sinh^2 y$.*
  Solution: (i) For $z = i$, $|i| = 1$ and $\mathrm{Arg}(i) = \frac{\pi}{2}$, so $\mathrm{Ln}(i) = \ln 1 + i\frac{\pi}{2} = \boxed{i\frac{\pi}{2}}$.
  (ii) Since $\sin(x+iy) = \sin x \cosh y + i \cos x \sinh y$:
  $|\sin z|^2 = \sin^2 x \cosh^2 y + \cos^2 x \sinh^2 y = \sin^2 x(1+\sinh^2 y) + (1-\sin^2 x)\sinh^2 y = \boxed{\sin^2 x + \sinh^2 y}$

- **Problem 5 (2014–15 Q1c, 2016–17 Q2b):** *Show that $\mathrm{Log}(-1+i)^2 \neq 2\,\mathrm{Log}(-1+i)$ (and similarly for $z = -1 + \sqrt{3}i$).*
  Solution: Let $z = -1 + i$. Then $|z| = \sqrt{2}$ and $\mathrm{Arg}(z) = \frac{3\pi}{4} \in (-\pi, \pi]$:

$$2\,\mathrm{Log}(-1+i) = 2\left(\ln\sqrt{2} + i\frac{3\pi}{4}\right) = \ln 2 + i\frac{3\pi}{2}$$

  On the other hand, $(-1+i)^2 = 1 - 2i - 1 = -2i$, which has modulus $|-2i| = 2$ and principal argument $\mathrm{Arg}(-2i) = -\frac{\pi}{2} \in (-\pi, \pi]$:

$$\mathrm{Log}(-1+i)^2 = \mathrm{Log}(-2i) = \ln 2 - i\frac{\pi}{2} \neq \ln 2 + i\frac{3\pi}{2}$$

  (For $z = -1 + \sqrt{3}i = 2e^{i2\pi/3}$, $z^2 = 4e^{i4\pi/3} = 4e^{-i2\pi/3}$, so $\mathrm{Log}(z^2) = \ln 4 - i\frac{2\pi}{3} \neq 2\,\mathrm{Log}(z) = \ln 4 + i\frac{4\pi}{3}$).

- **Problem 6 (2022–23 Q2c):** *If $\bar{z}_1 + i\bar{z}_2 = 0$ and $\arg(z_1 z_2) = \pi$, find $\arg(z_1)$.*
  Solution: Taking the complex conjugate of $\bar{z}_1 + i\bar{z}_2 = 0$ gives $z_1 - iz_2 = 0 \implies z_2 = \frac{z_1}{i} = -iz_1 = e^{-i\pi/2} z_1$. Thus $\arg(z_2) = \arg(z_1) - \frac{\pi}{2}$. Substituting into $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2) = \pi$:

$$\arg(z_1) + \left(\arg(z_1) - \frac{\pi}{2}\right) = \pi \implies 2\arg(z_1) = \frac{3\pi}{2} \implies \boxed{\arg(z_1) = \frac{3\pi}{4}}$$

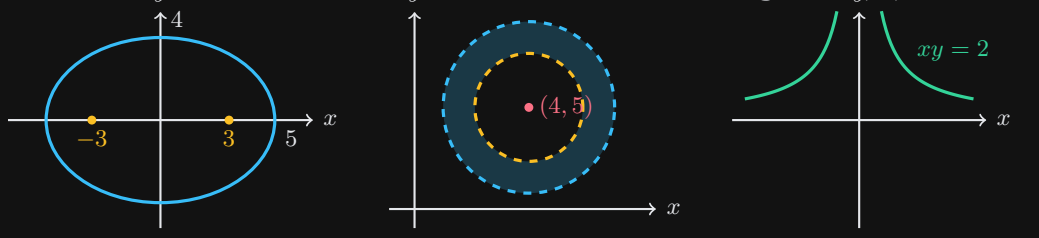
- **Problem 7 (2010–11 Q1c):** *Show that $\sin^{-1}(\csc\theta) = \frac{\pi}{2} + i\ln\left(\cot\frac{\theta}{2}\right)$.*
  Solution: Let $w = u + iv = \sin^{-1}(\csc\theta) \implies \sin(u+iv) = \sin u \cosh v + i\cos u \sinh v = \csc\theta$. Equating real and imaginary parts gives $\sin u \cosh v = \csc\theta > 1$ and $\cos u \sinh v = 0$. Since $\cosh v \geq 1$ and $\csc\theta > 1$, $\sinh v \neq 0$, so $\cos u = 0 \implies u = \frac{\pi}{2}$ ($\sin u = 1$). Then $\cosh v = \csc\theta \implies \sinh v = \pm\sqrt{\csc^2\theta - 1} = \cot\theta$. Thus $e^v = \cosh v + \sinh v = \csc\theta + \cot\theta = \frac{1+\cos\theta}{\sin\theta} = \cot\frac{\theta}{2} \implies v = \ln\left(\cot\frac{\theta}{2}\right)$, proving $\boxed{w = \frac{\pi}{2} + i\ln\left(\cot\frac{\theta}{2}\right)}$.

1.2 Standard Complex Loci & Regions in the Argand Plane

Exam Appearances: `2011-12 Q1a (8)`, `2012-13 Q1a (8)`, `2016-17 Q1a (5)`, `2024-25 Q1a (12)`.
Textbook Citations: M. R. Spiegel, *Schaum's Outline of Complex Variables*, Ch. 1, Sec. 1.7.

1. **Ellipse Locus ($|z-c_1|+|z-c_2| = 2a$ or $\leq 2a$):** Represents an ellipse (or closed elliptical disk) with foci at c_1 and c_2 ($c = \frac{1}{2}|c_1 - c_2| < a$), semi-major axis a, and semi-minor axis $b = \sqrt{a^2 - c^2}$:
   - $|z+2i| + |z-2i| \leq 6$ (`2011-12 Q1a-i`): Foci at $(0, \pm 2)$, $c = 2, a = 3, b = \sqrt{9-4} = \sqrt{5} \implies \boxed{\frac{x^2}{5} + \frac{y^2}{9} \leq 1}$.
   - $|z+1|+|z-1| \leq 4$ (`2012-13 Q1a-ii`): Foci at $(\pm 1, 0)$, $c = 1, a = 2, b = \sqrt{3} \implies \boxed{\frac{x^2}{4} + \frac{y^2}{3} \leq 1}$.
   - $|z-4i| + |z+4i| = 10$ (`2016-17 Q1a-ii`): Foci at $(0, \pm 4)$, $c = 4, a = 5, b = 3 \implies \boxed{\frac{x^2}{9} + \frac{y^2}{25} = 1}$.
   - $|z-3|+|z+3| = 10$ (`2024-25 Q1a-ii`): Foci at $(\pm 3, 0)$, $c = 3, a = 5, b = 4 \implies \boxed{\frac{x^2}{25} + \frac{y^2}{16} = 1}$.
2. **Circle & Annulus Loci:**
   - $2 < |z - 4 - 5i| < 3$ (`2012-13 Q1a-i`): Open circular annulus (ring) centered at $(4, 5)$ with inner radius $r_1 = 2$ and outer radius $r_2 = 3$.
   - $|z + i| \leq 3$ (`2016-17 Q1a-i`): Closed circular disk $x^2 + (y+1)^2 \leq 9$ centered at $(0, -1)$ of radius 3.
   - $z(\bar{z} + 2) = 3$ (`2024-25 Q1a-i`): Expanding $x^2 + y^2 + 2x + i2y = 3$ and equating real and imaginary parts gives $y = 0$ and $x^2 + 2x - 3 = 0 \implies (x+3)(x-1) = 0$, which represents the two points $(1, 0)$ and $(-3, 0)$ on the real axis. *(Note: If interpreted as $z\bar{z}+2\,\mathrm{Re}(z) = 3$, then $(x+1)^2 + y^2 = 4$, a circle of center $(-1, 0)$ and radius 2).*
3. **Hyperbolic Region ($\mathrm{Im}(z^2) \leq 4$, `2011-12 Q1a-ii`):** Since $z^2 = (x^2 - y^2) + i(2xy)$, $\mathrm{Im}(z^2) \leq 4 \implies \boxed{xy \leq 2}$ (the region between and below the two branches of the rectangular hyperbola $xy = 2$).

(a) Ellipse $|z-3| + |z+3| = 10$ **(b) Annulus $2 < |z - z_0| < 3$** **(c) Region $\mathrm{Im}(z^2) = 2xy \leq 4$**



1.3 Branch Points & Riemann Surface of $f(z) = \ln z$

Exam Appearances: `2018-19 Q1a (11)`, `2020-21 Q2a (10)`.
Textbook Citations: J. W. Brown & R. V. Churchill, *Complex Variables and Applications*, Ch. 3, Sec. 31–32 & Ch. 8; M. R. Spiegel, *Schaum's Outline of Complex Variables*, Ch. 2 & Ch. 10.

1. Definitions: Branch, Branch Cut, Branch Point, and Riemann Surface

- **Branch of a Multiple-Valued Function:** Any single-valued function $F(z)$ that is analytic in some domain and assumes one of the values of a multiple-valued function $f(z)$ at each point of that domain.
- **Branch Cut:** A line or curve (typically a ray from a branch point to ∞) introduced to prevent a point z from making a complete circuit around a branch point, thereby keeping a chosen branch single-valued and analytic.
- **Branch Point:** A point z_0 such that a multiple-valued function $f(z)$ does not return to its original value when the variable z traverses a small simple closed circuit C surrounding z_0 once.

2. Proof that $f(z) = \ln z$ Has a Branch Point at $z = 0$

Write z in polar coordinates around the origin $z = 0$ as $z = re^{i\theta}$ ($r > 0$). Then:

$$f(z) = \ln z = \ln r + i\theta$$

Let z start at a point $z_1 = re^{i\theta_1}$ and traverse a complete counterclockwise circle of radius r centered at $z = 0$. As z completes one full revolution back to the same geometric point z_1, the modulus r remains unchanged, while the argument θ increases continuously from θ_1 to $\theta_1 + 2\pi$. The value of $f(z)$ becomes:

$$\ln r + i(\theta_1 + 2\pi) = (\ln r + i\theta_1) + 2\pi i \neq \ln r + i\theta_1$$

Because $f(z) = \ln z$ increases by $2\pi i$ and fails to return to its initial value after a closed circuit around $z = 0$, $\boxed{z = 0 \text{ is a branch point of } f(z) = \ln z}$ (and similarly $z = \infty$ is also a branch point).

3. Riemann Surface for $f(z) = \ln z$ (2020-21 Q2a)

To make $f(z) = \ln z = \ln r + i(\theta + 2n\pi)$ ($n \in \mathbb{Z}$) single-valued and continuous everywhere without restricting θ, Bernhard Riemann replaced the single z-plane with an infinite helicoidal stack of cut complex planes called **Riemann sheets** (S_n for $-\infty < n < \infty$, where $(2n-1)\pi < \theta \leq (2n+1)\pi$ on sheet S_n):

- Each sheet S_n is cut along the negative real axis (the branch cut $\theta = \pi$).
- The upper edge of the slit on sheet S_n is joined smoothly to the lower edge of the slit on sheet S_{n+1} for every integer n, forming a continuous infinite spiral ramp (like a multi-level parking garage) around the vertical axis at $z = 0$.
- Each revolution around $z = 0$ carries z smoothly from sheet S_n to sheet S_{n+1}, so $f(z) = \ln z$ becomes a **single-valued analytic function** on the entire Riemann surface.

2 Analyticity, Cauchy-Riemann Equations & Harmonic Functions

2.1 Analytic, Entire & Harmonic Functions, and Cauchy-Riemann Derivations

Exam Appearances: `2010-11 Q2a (18)`, `2011-12 Q1b,c,d`, `2012-13 Q1b,c`, `2013-14 Q1a,b`, `2014-15 Q1a`, `2016-17 Q1d, Q2a`, `2017-18 Q2a (12)`, `2018-19 Q2a,b`, `2021-22 Q2a (20)`, `2022-23 Q1c (11)`, `2023-24 Q2a,b (25)`, `2024-25 Q2b,c`.
Textbook Citations: J. W. Brown & R. V. Churchill, *Complex Variables and Applications*, Ch. 2, Sec. 18–26; M. R. Spiegel, *Schaum's Outline of Complex Variables*, Ch. 3.

1. Definitions: Analytic Function, Entire Function, and Harmonic Function

- **Analytic (Holomorphic / Regular) Function:** A complex-valued function $f(z)$ is said to be **analytic at a point** z_0 if its derivative $f'(z)$ exists not only at z_0 itself, but at *every point z in some ε-neighborhood $|z - z_0| < \varepsilon$* of z_0. It is analytic in a domain D if it is analytic at every point in D.
- **Entire Function:** A function $f(z)$ that is analytic at **every point in the entire finite complex plane** $\mathbb{C}$ (e.g., polynomials, e^z, $\sin z$, $\cos z$, $-ie^{iz}$).
- **Harmonic Function & Harmonic Conjugate:** A real-valued function $u(x, y)$ having continuous first and second partial derivatives in a domain D is called **harmonic** in D if it satisfies **Laplace's equation**:

$$\boxed{\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0} \tag{3}$$

If $f(z) = u(x, y) + iv(x, y)$ is analytic in D, both u and v are harmonic in D, and $v(x, y)$ is called the **harmonic conjugate** of $u(x, y)$, found via exact differential integration of $dv = \frac{\partial v}{\partial x}dx + \frac{\partial v}{\partial y}dy = -\frac{\partial u}{\partial y}dx + \frac{\partial u}{\partial x}dy$ (or directly via the **Milne-Thomson method**: $f'(z) = u_x(z, 0) - iu_y(z, 0)$).

2. Derivation of the Cauchy-Riemann Equations (Cartesian Form)

Theorem (Necessary & Sufficient Conditions for Analyticity): Let $f(z) = u(x, y) + iv(x, y)$ be defined throughout some neighborhood of $z = x + iy$. Then $f(z)$ is analytic in a domain D **if and only if** the first-order partial derivatives u_x, u_y, v_x, v_y exist, are **continuous** in D, and satisfy the **Cauchy-Riemann (C-R) equations**:

$$\boxed{\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \qquad \text{and} \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}} \tag{4}$$

Derivation (Necessary Condition): Suppose $f'(z)$ exists at $z = x + iy$. With $\Delta z = \Delta x + i\Delta y$:

$$f'(z) = \lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = \lim_{\substack{\Delta x \to 0 \\ \Delta y \to 0}} \frac{[u(x + \Delta x, y + \Delta y) - u(x, y)] + i\,[v(x + \Delta x, y + \Delta y) - v(x, y)]}{\Delta x + i\Delta y}$$

Since $f'(z)$ exists, the limit must be unique regardless of the path along which $\Delta z \to 0$:

1. **Path 1 (Along a line parallel to the real axis: $\Delta y = 0, \Delta x \to 0$):**

$$f'(z) = \lim_{\Delta x \to 0} \left[\frac{u(x + \Delta x, y) - u(x, y)}{\Delta x} + i \frac{v(x + \Delta x, y) - v(x, y)}{\Delta x} \right] = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \tag{5}$$

2. **Path 2 (Along a line parallel to the imaginary axis: $\Delta x = 0, \Delta y \to 0$):**

$$f'(z) = \lim_{\Delta y \to 0} \left[\frac{u(x, y + \Delta y) - u(x, y)}{i\Delta y} + i \frac{v(x, y + \Delta y) - v(x, y)}{i\Delta y} \right] = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y} \tag{6}$$

Equating the real and imaginary parts of Equations (5) and (6) directly gives:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \qquad \text{and} \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Sufficiency Proof Outline: If u_x, u_y, v_x, v_y are continuous and satisfy the C-R equations, then by the total differential theorem:

$$\Delta u = u_x \Delta x + u_y \Delta y + \varepsilon_1 \Delta x + \varepsilon_2 \Delta y, \qquad \Delta v = v_x \Delta x + v_y \Delta y + \eta_1 \Delta x + \eta_2 \Delta y$$

Substituting $u_y = -v_x$ and $v_y = u_x$ into $\Delta f = \Delta u + i\Delta v$ gives $\Delta f = (u_x + iv_x)(\Delta x + i\Delta y) + \text{error}$, so $\lim_{\Delta z \to 0} \frac{\Delta f}{\Delta z} = u_x + iv_x = f'(z)$ exists and is continuous.

3. Derivation of Cauchy-Riemann Equations in Polar Form (`2021-22 Q2a`, `2022-23 Q1c`, `2023-24 Q2b`)

Substitute $x = r\cos\theta$ and $y = r\sin\theta$ into $u(x, y)$ and $v(x, y)$. By the chain rule:

$$\begin{aligned}
\frac{\partial u}{\partial r} &= \frac{\partial u}{\partial x}\cos\theta + \frac{\partial u}{\partial y}\sin\theta, & \frac{\partial u}{\partial \theta} &= -\frac{\partial u}{\partial x} r\sin\theta + \frac{\partial u}{\partial y} r\cos\theta \\
\frac{\partial v}{\partial r} &= \frac{\partial v}{\partial x}\cos\theta + \frac{\partial v}{\partial y}\sin\theta, & \frac{\partial v}{\partial \theta} &= -\frac{\partial v}{\partial x} r\sin\theta + \frac{\partial v}{\partial y} r\cos\theta
\end{aligned}$$

Using the Cartesian C-R equations ($u_x = v_y$ and $u_y = -v_x$) in the equations for $\frac{\partial u}{\partial r}$ and $\frac{\partial u}{\partial \theta}$:

$$\begin{aligned}
\frac{\partial u}{\partial r} &= \frac{\partial v}{\partial y}\cos\theta - \frac{\partial v}{\partial x}\sin\theta = \frac{1}{r}\left(-\frac{\partial v}{\partial x} r\sin\theta + \frac{\partial v}{\partial y} r\cos\theta\right) = \frac{1}{r}\frac{\partial v}{\partial \theta} \\
\frac{\partial u}{\partial \theta} &= -\frac{\partial v}{\partial y} r\sin\theta - \frac{\partial v}{\partial x} r\cos\theta = -r\left(\frac{\partial v}{\partial x}\cos\theta + \frac{\partial v}{\partial y}\sin\theta\right) = -r\frac{\partial v}{\partial r}
\end{aligned}$$

Thus, the **Polar Form of the Cauchy-Riemann Equations** and the polar derivative $f'(z)$ are:

$$\boxed{\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta} \qquad \text{and} \qquad \frac{1}{r}\frac{\partial u}{\partial \theta} = -\frac{\partial v}{\partial r}}, \qquad \boxed{f'(z) = e^{-i\theta}\left(\frac{\partial u}{\partial r} + i\frac{\partial v}{\partial r}\right)} \tag{7}$$

4. Classic Short Proofs on C-R Equations, Differentiability & Analyticity

- **Counterexample 1 (`2011-12 Q1b`, `2013-14 Q1a`, `2024-25 Q2b`):**

$$f(z) = \begin{cases} \dfrac{(\bar{z})^2}{z}, & z \neq 0 \\ 0, & z = 0 \end{cases}$$

1. *Continuity at $z = 0$:* In polar form, for $z = re^{i\theta} \neq 0$, $|f(z) - f(0)| = \left|\frac{r^2 e^{-i2\theta}}{re^{i\theta}}\right| = r = |z| \to 0$ as $z \to 0$. Thus $\lim_{z\to 0} f(z) = 0 = f(0)$, so $f(z)$ is **continuous** at $z = 0$.
2. *C-R Equations at $z = 0$:* Since $f(z) = \frac{(x-iy)^3}{x^2+y^2} = \frac{x^3-3xy^2}{x^2+y^2} + i\frac{y^3-3x^2y}{x^2+y^2}$, along the axes $u(x, 0) = x, u(0, y) = 0, v(x, 0) = 0, v(0, y) = y$. Hence at $(0, 0)$:

$$u_x(0,0) = \lim_{x\to 0}\frac{x-0}{x} = 1, \quad v_y(0,0) = \lim_{y\to 0}\frac{y-0}{y} = 1, \quad u_y(0,0) = 0 = -v_x(0,0)$$

Thus the **Cauchy-Riemann equations are satisfied at** $(0, 0)$.

3. *Non-Differentiability at $z = 0$:* Approaching $z = 0$ along the radial line $y = mx$ ($z = re^{i\theta}$):

$$f'(0) = \lim_{z\to 0} \frac{f(z) - f(0)}{z} = \lim_{z\to 0} \left(\frac{\bar{z}}{z}\right)^2 = \lim_{x\to 0} \left(\frac{1 - im}{1 + im}\right)^2 = e^{-i4\theta}$$

Since this limit depends on the slope m (e.g., $+1$ for $m = 0$ vs. -1 for $m = 1$), $f'(0)$ **does not exist**.

- **Counterexample 2 (`2012-13 Q1c`, `2013-14 Q1b`, `2016-17 Q2a-i`, `2018-19 Q2a`):** *Functions differentiable only at isolated points / nowhere analytic:*
  Shortcut (Wirtinger Derivative $\frac{\partial f}{\partial \bar{z}} = 0$): A function with continuous partials satisfies the C-R equations at z if and only if $\frac{\partial f}{\partial \bar{z}} = 0$.
  - For $f(z) = i\bar{z}^2 = 2xy + i(x^2 - y^2)$ (`2013-14 Q1b`, `2016-17 Q2a-i`): $u_x = 2y, v_y = -2y$ and $u_y = 2x, -v_x = -2x$. The C-R equations hold **only at** $(0, 0)$. Thus $f(z)$ is differentiable **only at** $z = 0$ and is **analytic nowhere**.
  - For $f(z) = z\bar{z} + \bar{z} = (x^2 + y^2 + x) - iy$ (`2012-13 Q1c`): $u_x = 2x + 1, v_y = -1$ and $u_y = 2y, -v_x = 0$. C-R equations hold **only at** $(-1, 0)$ ($z = -1$). Thus $f(z)$ is differentiable only at $z = -1$ and **analytic nowhere**.
  - For $f(z) = z^2\bar{z} = (x^3 + xy^2) + i(x^2y + y^3)$ (`2018-19 Q2a`): $u_x = 3x^2 + y^2$ and $v_y = x^2 + 3y^2$, which are equal only when $x^2 = y^2$; and $u_y = 2xy = -v_x = -2xy \implies xy = 0$. Together, C-R holds only at $z = 0$ (nowhere in $z \neq 0$, so nowhere analytic).

- **Entire Function Proofs (`2010-11 Q2a`, `2014-15 Q1a`, `2016-17 Q2a-ii`, `2024-25 Q2c`):**
  - For $f(z) = -ie^{iz} = -ie^{-y}(\cos x + i \sin x) = e^{-y} \sin x - ie^{-y} \cos x$: $u_x = e^{-y} \cos x = v_y$ and $u_y = -e^{-y} \sin x = -v_x$ for all $(x, y) \in \mathbb{R}^2$, with continuous partials everywhere. Hence $f(z) = -ie^{iz}$ is an **entire function**.
  - For $f(z) = e^{z^2} = e^{x^2-y^2}(\cos 2xy + i \sin 2xy)$: C-R equations hold everywhere in $\mathbb{C}$ (entire), and $|e^{z^2}| = e^{\mathrm{Re}(z^2)} = e^{x^2-y^2} \leq e^{x^2+y^2} = e^{|z|^2}$ since $-y^2 \leq y^2$ (`2014-15 Q1a`). Meanwhile, $g(z) = e^{|z|^2} = e^{x^2+y^2} + i0$ satisfies $u_x = 2xe^{x^2+y^2} = v_y = 0$ and $u_y = 2ye^{x^2+y^2} = -v_x = 0$ only at $z = 0$, so $e^{|z|^2}$ is differentiable only at $z = 0$ and nowhere analytic.

3 Complex Integration, Cauchy's Theorems, Series & Residue Calculus

3.1 ML-Inequality, Cauchy's Theorem, Cauchy's Integral Formula & Complex Green's Theorem

Exam Appearances: `2011-12 Q3b`, `2014-15 Q2a`, `2017-18 Q3a,b (18)`, `2018-19 Q3a,b, Q4a,b`, `2019-20 Q3b`, `2020-21 Q3b`, `2022-23 Q2b`, `2023-24 Q3b,c`, `2024-25 Q3b,c`.
Textbook Citations: J. W. Brown & R. V. Churchill, *Complex Variables and Applications*, Ch. 4, Sec. 43–54; M. R. Spiegel, *Schaum's Outline of Complex Variables*, Ch. 4, Sec. 4.4–4.7.

1. Statement and Proof of the ML-Inequality (`2017-18 Q3a`, `2023-24 Q3b`)

Theorem: Let C denote a contour of finite arc length L, and suppose that $f(z)$ is piecewise continuous on C. If there exists a non-negative constant M such that $|f(z)| \leq M$ for all points z on C, then:

$$\boxed{\left|\int_C f(z)\,dz\right| \leq ML} \tag{8}$$

Proof: Let C be parametrized by $z(t)$ ($a \leq t \leq b$), or partitioned into n sub-arcs $\Delta z_k = z_k - z_{k-1}$ of chord lengths $|\Delta z_k|$ with sample points ζ_k on each sub-arc. By the generalized triangle inequality for Riemann sums:

$$\left|\sum_{k=1}^{n} f(\zeta_k)\Delta z_k\right| \leq \sum_{k=1}^{n} |f(\zeta_k)|\,|\Delta z_k| \leq M\sum_{k=1}^{n}|\Delta z_k|$$

Taking the limit as the norm of the partition approaches zero ($\max|\Delta z_k| \to 0$), the left-hand side converges to $\left|\int_C f(z)\,dz\right|$ and $\sum_{k=1}^{n}|\Delta z_k|$ converges to the total arc length $L = \int_C |dz| = \int_a^b |z'(t)|\,dt$, proving $\left|\int_C f(z)\,dz\right| \leq ML$.

Standard Exam Applications of the ML-Inequality:

- **Example 1 (`2023-24 Q3b`, `2024-25 Q3b`):** On the first-quadrant circular arc $C : |z| = 2$ from $z = 2$ to $z = 2i$, the arc length is $L = \frac{1}{4}(2\pi \cdot 2) = \pi$. By the triangle inequalities, $|z^3 - 1| \geq \big||z|^3 - 1\big| = 2^3 - 1 = 7$, while $|z + 4| \leq |z| + 4 = 6$ (or $|z + 1| \leq |z| + 1 = 3$). Thus $M = \frac{6}{7}$ (or $\frac{3}{7}$), giving $\boxed{\left|\int_C \frac{z+4}{z^3-1}dz\right| \leq \frac{6\pi}{7}}$ and $\boxed{\left|\int_C \frac{z+1}{z^3-1}dz\right| \leq \frac{3\pi}{7}}$.
- **Example 2 (`2017-18 Q3b`):** On the first-quadrant arc of $|z| = 2$, $L = \pi$ and $|z^2 - 1| \geq |z|^2 - 1 = 4 - 1 = 3 \implies M = \frac{1}{3}$, so $\boxed{\left|\int_C \frac{dz}{z^2-1}\right| \leq \frac{\pi}{3}}$.
- **Example 3 (`2011-12 Q3b`, `2014-15 Q2a`):** On the upper semicircle $C_R : |z| = R\,(R > 2)$, $L = \pi R$, $|2z^2 - 1| \leq 2|z|^2 + 1 = 2R^2 + 1$, and $|z^4 + 5z^2 + 4| = |(z^2 + 1)(z^2 + 4)| \geq (|z|^2 - 1)(|z|^2 - 4) = (R^2 - 1)(R^2 - 4)$, giving $\boxed{\left|\int_{C_R} \frac{2z^2-1}{z^4+5z^2+4}dz\right| \leq \frac{\pi R(2R^2+1)}{(R^2-1)(R^2-4)}}$.

2. Statement and Proof of Cauchy's Integral Theorem (`2018-19 Q3a`)

Theorem: If a function $f(z)$ is analytic at all points interior to and on a simple closed contour C, and if its derivative $f'(z)$ is continuous there, then:

$$\boxed{\oint_C f(z)\,dz = 0} \tag{9}$$

Proof (via Green's Theorem): Write $f(z) = u(x,y) + iv(x,y)$ and $dz = dx + i\,dy$:

$$\oint_C f(z)\,dz = \oint_C (u+iv)(dx+i\,dy) = \oint_C (u\,dx - v\,dy) + i\oint_C (v\,dx + u\,dy)$$

Since $f'(z)$ is continuous inside and on C, the partial derivatives u_x, u_y, v_x, v_y are continuous in the closed region R bounded by C. Applying Green's theorem in the plane ($\oint_C (P\,dx + Q\,dy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy$) to both real line integrals:

$$\oint_C f(z)\,dz = \iint_R \left(-\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right) dx\,dy + i\iint_R \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y}\right) dx\,dy$$

Because $f(z)$ is analytic in R, the Cauchy-Riemann equations $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$ hold throughout R. Both integrands vanish identically, proving $\oint_C f(z)\,dz = 0$.

3. Cauchy's Integral Formula & Fundamental Loop Integral

Cauchy's Integral Formula: Let $f(z)$ be analytic everywhere inside and on a simple closed contour C, taken in the positive (counterclockwise) sense. If z_0 is any point interior to C, then:

$$\boxed{f(z_0) = \frac{1}{2\pi i}\oint_C \frac{f(z)}{z - z_0}\,dz} \quad \text{and} \quad \boxed{f^{(n)}(z_0) = \frac{n!}{2\pi i}\oint_C \frac{f(z)}{(z-z_0)^{n+1}}\,dz} \quad (n = 1, 2, \dots) \tag{10}$$

Fundamental Loop Integral (`2018-19 Q4a`): For any point a interior to a simple closed contour C, deform C to a circle $z - a = re^{i\theta}$ ($0 \le \theta \le 2\pi, dz = ire^{i\theta}d\theta$):

$$\oint_C \frac{dz}{(z-a)^n} = \int_0^{2\pi} \frac{ire^{i\theta}\,d\theta}{r^n e^{in\theta}} = ir^{1-n}\int_0^{2\pi} e^{i(1-n)\theta}\,d\theta = \boxed{\begin{cases} 2\pi i, & n = 1 \\ 0, & n \neq 1 \text{ (e.g., } n = 5 \implies 0) \end{cases}}$$

4. Complex Form of Green's Theorem & Complex Differential Operators (`2018-19 Q3b, Q4b`)

- **Complex Form of Green's Theorem (`2018-19 Q3b`):** Let $P(z,\bar{z})$ and $Q(z,\bar{z})$ be continuous with continuous partial derivatives in a simply connected region R bounded by a simple closed curve C. Then:

$$\boxed{\oint_C [P(z,\bar{z})\,dz + Q(z,\bar{z})\,d\bar{z}] = 2i\iint_R \left(\frac{\partial P}{\partial \bar{z}} - \frac{\partial Q}{\partial z}\right) dx\,dy} \tag{11}$$

  In particular, for a single function $F(z,\bar{z})$, $\boxed{\oint_C F(z,\bar{z})\,dz = 2i\iint_R \frac{\partial F}{\partial \bar{z}}\,dx\,dy}$, and for $F(z,\bar{z}) = \bar{z}$ ($\frac{\partial \bar{z}}{\partial \bar{z}} = 1$), $\boxed{\oint_C \bar{z}\,dz = 2i \times (\text{Area enclosed by } C)}$ (used in `2023-24 Q3a` and `2024-25 Q3a`, where the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is πab, giving $\oint_C \bar{z}\,dz = 2\pi i ab$).
- **Complex Gradient, Divergence, and Curl Operators (`2018-19 Q4b`):** Defining the complex del operator $\nabla \equiv \frac{\partial}{\partial x} + i\frac{\partial}{\partial y} = 2\frac{\partial}{\partial \bar{z}}$ for $A(x,y) = P(x,y) + iQ(x,y)$:

  $$\text{grad}\,A = \nabla A = \left(\frac{\partial P}{\partial x} - \frac{\partial Q}{\partial y}\right) + i\left(\frac{\partial Q}{\partial x} + \frac{\partial P}{\partial y}\right), \quad \text{div}\,A = \text{Re}(\bar{\nabla}A) = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}, \quad \text{curl}\,A = \text{Im}(\bar{\nabla}A) = \frac{\partial Q}{\partial x} -$$

3.2 Laurent's Theorem, Singularities, Residue Theorem & Bilinear Mappings

Exam Appearances: `2010-11 Q3a,b, Q4a, 2013-14 Q4, 2014-15 Q4, 2018-19 Q1c, Q4c, 2023-24 Q4b,c, 2024-25 Q4a,b,c.`
Textbook Citations: J. W. Brown & R. V. Churchill, *Complex Variables and Applications*, Ch. 5, Sec. 60–62 & Ch. 6, Sec. 68–76; M. R. Spiegel, *Schaum's Outline of Complex Variables*, Ch. 6 & Ch. 7.

1. Statement and Proof of Laurent's Theorem (`2010-11 Q3a, 2023-24 Q4b, 2024-25 Q4a`)

Theorem: Let $f(z)$ be analytic throughout an annular domain $R_1 < |z - z_0| < R_2$ centered at z_0, and let C be any positively oriented simple closed contour around z_0 lying completely inside this annulus. Then at each point z in the annulus, $f(z)$ has the series representation:

$$\boxed{f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z - z_0)^n} = \sum_{n=-\infty}^{\infty} c_n (z - z_0)^n} \tag{12}$$

where the coefficients are given by:

$$a_n = \frac{1}{2\pi i} \oint_C \frac{f(w)}{(w - z_0)^{n+1}}\, dw \quad (n = 0, 1, 2, \dots), \qquad b_n = \frac{1}{2\pi i} \oint_C \frac{f(w)}{(w - z_0)^{-n+1}}\, dw \quad (n = 1, 2, \dots) \tag{13}$$

Proof: Let z be any point in the annulus $R_1 < |z - z_0| < R_2$. Choose two concentric circles C_1 ($|w - z_0| = r_1$) and C_2 ($|w - z_0| = r_2$) inside the annulus such that $R_1 < r_1 < |z - z_0| < r_2 < R_2$. By Cauchy's integral formula for a multiply connected domain:

$$f(z) = \frac{1}{2\pi i} \oint_{C_2} \frac{f(w)}{w - z}\, dw - \frac{1}{2\pi i} \oint_{C_1} \frac{f(w)}{w - z}\, dw$$

- On C_2, $|z - z_0| < |w - z_0|$, so $\frac{1}{w-z} = \frac{1}{(w-z_0)-(z-z_0)} = \sum_{n=0}^{\infty} \frac{(z-z_0)^n}{(w-z_0)^{n+1}}$. Integrating term-by-term gives $\sum_{n=0}^{\infty} a_n (z - z_0)^n$.
- On C_1, $|w - z_0| < |z - z_0|$, so $-\frac{1}{w-z} = \frac{1}{(z-z_0)-(w-z_0)} = \sum_{n=1}^{\infty} \frac{(w-z_0)^{n-1}}{(z-z_0)^n}$. Integrating term-by-term gives $\sum_{n=1}^{\infty} \frac{b_n}{(z-z_0)^n}$.

By the principle of deformation of contours, C_1 and C_2 can both be replaced by any simple closed contour C in the annulus encircling z_0, completing the proof.

2. Classification of Singularities & Residue Formulas

A point z_0 is a **singular point (singularity)** of $f(z)$ if $f(z)$ fails to be analytic at z_0, but is analytic at some point in every neighborhood of z_0. It is an **isolated singularity** if $f(z)$ is analytic everywhere in $0 < |z - z_0| < \varepsilon$:

1. **Removable Singularity:** All negative-power coefficients in the Laurent series about z_0 vanish ($b_n = 0$ for all $n \geq 1$), or equivalently $\lim_{z \to z_0} (z - z_0) f(z) = 0$ (e.g., $f(z) = \frac{\sin z}{z}$ at $z = 0$). Here $\operatorname{Res}_{z=z_0} f(z) = 0$.
2. **Pole of Order m:** The principal part of the Laurent series has a finite number of terms, ending at $b_m \neq 0$ with $b_{m+1} = b_{m+2} = \cdots = 0$ (when $m = 1$, it is a **simple pole**).
3. **Essential Singularity:** Infinitely many negative-power coefficients b_n are non-zero (e.g., $z^5 e^{1/z^2}$ at $z = 0$, or $\frac{z-1}{3z^2} \cos\left(\frac{1}{z-1}\right)$ at $z = 1$). Here $\operatorname{Res}_{z=z_0} f(z) = b_1$ is found by expanding the series directly.

Master Residue Formulas:

$$\text{General Definition:} \quad \text{Res}_{z=z_0} f(z) = b_1 = \frac{1}{2\pi i} \oint_C f(z)\, dz \tag{14}$$

$$\text{Simple Pole } (m=1): \quad \text{Res}_{z=z_0} f(z) = \lim_{z \to z_0} (z - z_0) f(z) = \frac{P(z_0)}{Q'(z_0)} \quad \left(\text{where } f = \frac{P}{Q}, Q(z_0) = 0, Q'(z_0) \neq 0 \right) \tag{15}$$

$$\text{Pole of Order } m: \quad \text{Res}_{z=z_0} f(z) = \frac{1}{(m-1)!} \lim_{z \to z_0} \frac{d^{m-1}}{dz^{m-1}} \left[(z - z_0)^m f(z) \right] \tag{16}$$

3. Cauchy's Residue Theorem & Bilinear (Möbius) Transformation

- **Cauchy's Residue Theorem:** Let C be a simple closed contour, described in the positive (counterclockwise) sense. If a function $f(z)$ is analytic inside and on C except for a finite number of singular points $z_1, z_2, \ldots, z_n$ interior to C, then:

$$\boxed{\oint_C f(z)\, dz = 2\pi i \sum_{k=1}^{n} \text{Res}_{z=z_k} f(z)} \tag{17}$$

- **Bilinear (Linear Fractional / Möbius) Transformation & Cross-Ratio:** A mapping of the form $w = \frac{az+b}{cz+d}$ ($ad - bc \neq 0$), which always maps circles and lines onto circles and lines. The unique bilinear transformation mapping three distinct points (z_1, z_2, z_3) onto (w_1, w_2, w_3) is given by the **invariance of the cross-ratio**:

$$\boxed{\frac{(w - w_1)(w_2 - w_3)}{(w - w_3)(w_2 - w_1)} = \frac{(z - z_1)(z_2 - z_3)}{(z - z_3)(z_2 - z_1)}} \tag{18}$$

Part II

Section B: Vector Analysis & Vector Calculus (Definitions, Theorems, Proofs & Short Questions)

4 Vector Algebra, Linear Dependence, Coplanarity & Reciprocal Vectors

4.1 Linear Dependence/Independence, Coplanarity & Reciprocal Vector Proofs

Exam Appearances: `2010-11 Q5a,b`, `2011-12 Q5a,c`, `2012-13 Q5a,b,c`, `2013-14 Q5b,c`, `2014-15 Q5a,c`, `2016-17 Q5a,c`, `2017-18 Q5b`, `2018-19 Q5a,b`, `2019-20 Q5a, Q7a`, `2020-21 Q5a,b,c, Q6b`, `2021-22 Q5a,b`, `2022-23 Q5a,b`, `2023-24 Q5a`, `2024-25 Q5a`.
Textbook Citations: M. R. Spiegel, *Schaum's Outline of Vector Analysis* (2nd Ed.), Ch. 1 & Ch. 2.

1. Definitions: Linear Dependence, Linear Independence, Coplanarity & Collinearity

- **Linearly Dependent and Independent Vectors (`2024-25 Q5a`):** A set of m vectors $\vec{A}_1, \vec{A}_2, \ldots, \vec{A}_m$ is said to be **linearly dependent** if there exist scalars $c_1, c_2, \ldots, c_m$, *not all zero*, such that:

$$c_1\vec{A}_1 + c_2\vec{A}_2 + \cdots + c_m\vec{A}_m = \vec{0}$$

  If this vector equation holds **only** when $c_1 = c_2 = \cdots = c_m = 0$ (the trivial solution), the vectors are said to be **linearly independent**.
  - *Determinant Test for Three 3D Vectors:* Three vectors $\vec{a}, \vec{b}, \vec{c}$ in $\mathbb{R}^3$ are linearly independent (non-coplanar) if and only if their scalar triple product is non-zero: $\boxed{[\vec{a}\,\vec{b}\,\vec{c}] = \vec{a} \cdot (\vec{b} \times \vec{c}) \neq 0}$.
  - *Note on Four 3D Vectors:* Any four vectors in 3D space are always linearly dependent.
- **Condition for Coplanarity of Four Points (`2013-14 Q5b`, `2017-18 Q5b`, `2019-20 Q5a`):** Four points A, B, C, D with position vectors $\vec{r}_1, \vec{r}_2, \vec{r}_3, \vec{r}_4$ are coplanar if and only if the three edge vectors $\vec{AB} = \vec{r}_2 - \vec{r}_1$, $\vec{AC} = \vec{r}_3 - \vec{r}_1$, and $\vec{AD} = \vec{r}_4 - \vec{r}_1$ lie in the same plane (so the volume of the tetrahedron $V = \frac{1}{6}\left|[\vec{AB}, \vec{AC}, \vec{AD}]\right|$ is zero):

$$\boxed{[\vec{r}_2 - \vec{r}_1,\ \vec{r}_3 - \vec{r}_1,\ \vec{r}_4 - \vec{r}_1] = 0} \iff \boxed{\alpha\vec{r}_1 + \beta\vec{r}_2 + \gamma\vec{r}_3 + \delta\vec{r}_4 = \vec{0} \text{ with } \alpha + \beta + \gamma + \delta = 0} \quad (19)$$

  *(Similarly, three points $\vec{r}_1, \vec{r}_2, \vec{r}_3$ are **collinear** if and only if $\alpha\vec{r}_1 + \beta\vec{r}_2 + \gamma\vec{r}_3 = \vec{0}$ with $\alpha + \beta + \gamma = 0$).*
- **How Many Planes Can Be Constructed Using Two Points? (`2022-23 Q5b`): Infinitely many planes** can be constructed passing through two given distinct points P and Q. *Justification:* Two distinct points uniquely determine a single straight line PQ, which has only one direction vector $\vec{PQ}$. A unique plane requires a normal vector $\vec{n}$ perpendicular to two non-parallel direction vectors (requiring at least three non-collinear points). Any plane containing the line PQ can be rotated through 360° around the axis PQ (forming a "pencil of planes"), giving an infinite family of planes.

2. Proof That Any Vector $\vec{r}$ Is a Linear Combination of Three Non-Coplanar Vectors $\vec{a}, \vec{b}, \vec{c}$ (2012–13 Q5b, 2014–15 Q5a, 2016–17 Q5a)

Proof: Since $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar, $[\vec{a}\,\vec{b}\,\vec{c}] \neq 0$, and they span $\mathbb{R}^3$. Write:

$$\vec{r} = x\vec{a} + y\vec{b} + z\vec{c} \tag{20}$$

To find the unique scalars x, y, z:

1. Take the dot product of Equation (20) with $(\vec{b} \times \vec{c})$. Since $(\vec{b} \times \vec{c})$ is perpendicular to both $\vec{b}$ and $\vec{c}$, $\vec{b} \cdot (\vec{b} \times \vec{c}) = 0$ and $\vec{c} \cdot (\vec{b} \times \vec{c}) = 0$:

$$\vec{r} \cdot (\vec{b} \times \vec{c}) = x\,\vec{a} \cdot (\vec{b} \times \vec{c}) \implies x = \frac{[\vec{r}\,\vec{b}\,\vec{c}]}{[\vec{a}\,\vec{b}\,\vec{c}]}$$

2. Similarly, dotting Equation (20) with $(\vec{c} \times \vec{a})$ and $(\vec{a} \times \vec{b})$ yields $y = \frac{[\vec{a}\,\vec{r}\,\vec{c}]}{[\vec{a}\,\vec{b}\,\vec{c}]}$ and $z = \frac{[\vec{a}\,\vec{b}\,\vec{r}]}{[\vec{a}\,\vec{b}\,\vec{c}]}$.

Hence:

$$\boxed{\vec{r} = \frac{[\vec{r}\,\vec{b}\,\vec{c}]}{[\vec{a}\,\vec{b}\,\vec{c}]}\vec{a} + \frac{[\vec{a}\,\vec{r}\,\vec{c}]}{[\vec{a}\,\vec{b}\,\vec{c}]}\vec{b} + \frac{[\vec{a}\,\vec{b}\,\vec{r}]}{[\vec{a}\,\vec{b}\,\vec{c}]}\vec{c}} \tag{21}$$

3. Reciprocal System of Vectors & Expansion Proof (2018–19 Q5b, 2020–21 Q5c, 2021–22 Q5b)

Definition: Two sets of non-coplanar vectors $(\vec{a}, \vec{b}, \vec{c})$ and $(\vec{a}', \vec{b}', \vec{c}')$ are called **reciprocal systems of vectors** if $\vec{a} \cdot \vec{a}' = \vec{b} \cdot \vec{b}' = \vec{c} \cdot \vec{c}' = 1$ and $\vec{a}' \cdot \vec{b} = \vec{a}' \cdot \vec{c} = \vec{b}' \cdot \vec{a} = \vec{b}' \cdot \vec{c} = \vec{c}' \cdot \vec{a} = \vec{c}' \cdot \vec{b} = 0$, which is satisfied if and only if:

$$\boxed{\vec{a}' = \frac{\vec{b} \times \vec{c}}{[\vec{a}\,\vec{b}\,\vec{c}]}, \qquad \vec{b}' = \frac{\vec{c} \times \vec{a}}{[\vec{a}\,\vec{b}\,\vec{c}]}, \qquad \vec{c}' = \frac{\vec{a} \times \vec{b}}{[\vec{a}\,\vec{b}\,\vec{c}]}} \qquad \Big(\text{where } [\vec{a}\,\vec{b}\,\vec{c}] \neq 0\Big) \tag{22}$$

Proof that $\vec{r} = (\vec{r} \cdot \vec{a})\vec{a}' + (\vec{r} \cdot \vec{b})\vec{b}' + (\vec{r} \cdot \vec{c})\vec{c}'$ (2018–19 Q5b): Since $\vec{a}', \vec{b}', \vec{c}'$ are non-coplanar, write $\vec{r} = c_1\vec{a}' + c_2\vec{b}' + c_3\vec{c}'$. Dotting both sides with $\vec{a}$ gives:

$$\vec{r} \cdot \vec{a} = c_1(\vec{a}' \cdot \vec{a}) + c_2(\vec{b}' \cdot \vec{a}) + c_3(\vec{c}' \cdot \vec{a}) = c_1(1) + 0 + 0 = c_1$$

Similarly, dotting with $\vec{b}$ and $\vec{c}$ gives $c_2 = \vec{r} \cdot \vec{b}$ and $c_3 = \vec{r} \cdot \vec{c}$, proving:

$$\boxed{\vec{r} = (\vec{r} \cdot \vec{a})\vec{a}' + (\vec{r} \cdot \vec{b})\vec{b}' + (\vec{r} \cdot \vec{c})\vec{c}'} \tag{23}$$

4. Classic Vector Algebra Proofs & Vector Equations

- **Proof 1 (2011–12 Q5c, 2016–17 Q5c):** *Prove* $[\vec{a}\times\vec{p}, \vec{b}\times\vec{q}, \vec{c}\times\vec{r}] + [\vec{a}\times\vec{q}, \vec{b}\times\vec{r}, \vec{c}\times\vec{p}] + [\vec{a}\times\vec{r}, \vec{b}\times\vec{p}, \vec{c}\times\vec{q}] = 0$.
  Proof: Using $(\vec{A} \times \vec{B}) \cdot (\vec{C} \times \vec{D}) = [\vec{A}\vec{B}\vec{D}]\vec{C} - [\vec{A}\vec{B}\vec{C}]\vec{D}$:

$$[\vec{a} \times \vec{p}, \vec{b} \times \vec{q}, \vec{c} \times \vec{r}] = (\vec{a} \times \vec{p}) \cdot \Big\{(\vec{b} \times \vec{q}) \times (\vec{c} \times \vec{r})\Big\} = (\vec{a} \times \vec{p}) \cdot \Big\{[\vec{b}\vec{q}\vec{r}]\vec{c} - [\vec{b}\vec{q}\vec{c}]\vec{r}\Big\}$$
$$= [\vec{a}\vec{p}\vec{c}][\vec{b}\vec{q}\vec{r}] - [\vec{a}\vec{p}\vec{r}][\vec{b}\vec{q}\vec{c}] = -[\vec{a}\vec{c}\vec{p}][\vec{b}\vec{q}\vec{r}] + [\vec{a}\vec{r}\vec{p}][\vec{b}\vec{c}\vec{q}]$$

  Wait—it is even cleaner to expand $(\vec{a} \times \vec{p}) \times (\vec{b} \times \vec{q}) = [\vec{a}\vec{b}\vec{q}]\vec{p} - [\vec{p}\vec{b}\vec{q}]\vec{a}$:

$$T_1 = [\vec{a}\vec{b}\vec{q}][\vec{p}\vec{c}\vec{r}] - [\vec{p}\vec{b}\vec{q}][\vec{a}\vec{c}\vec{r}]$$
$$T_2 = [\vec{a}\vec{b}\vec{r}][\vec{q}\vec{c}\vec{p}] - [\vec{q}\vec{b}\vec{r}][\vec{a}\vec{c}\vec{p}]$$
$$T_3 = [\vec{a}\vec{b}\vec{p}][\vec{r}\vec{c}\vec{q}] - [\vec{r}\vec{b}\vec{p}][\vec{a}\vec{c}\vec{q}]$$

  Using the four-vector identity $[\vec{a}\vec{b}\vec{c}]\vec{d} = [\vec{d}\vec{b}\vec{c}]\vec{a} + [\vec{a}\vec{d}\vec{c}]\vec{b} + [\vec{a}\vec{b}\vec{d}]\vec{c}$, the six terms cancel in cyclic pairs, giving $\boxed{T_1 + T_2 + T_3 = 0}$.

- **Proof 2 (2022–23 Q5a):** *Express $(\vec{A}\times\vec{B})\times(\vec{C}\times\vec{D})$ in terms of $\vec{A},\vec{B}$ and find the volume of the parallelepiped with edges $\vec{B}\times\vec{C},\vec{C}\times\vec{A},\vec{A}\times\vec{B}$.*
  Proof: Let $\vec{U}=\vec{C}\times\vec{D}$. Then:

$$(\vec{A}\times\vec{B})\times\vec{U}=(\vec{A}\cdot\vec{U})\vec{B}-(\vec{B}\cdot\vec{U})\vec{A}=\boxed{[\vec{A}\vec{C}\vec{D}]\vec{B}-[\vec{B}\vec{C}\vec{D}]\vec{A}}$$

  For the parallelepiped with edges $\vec{B}\times\vec{C},\vec{C}\times\vec{A},\vec{A}\times\vec{B}$:

$$V=\left|(\vec{B}\times\vec{C})\cdot\left\{(\vec{C}\times\vec{A})\times(\vec{A}\times\vec{B})\right\}\right|=\left|(\vec{B}\times\vec{C})\cdot\left\{[\vec{C}\vec{A}\vec{B}]\vec{A}-\underbrace{[\vec{A}\vec{A}\vec{B}]}_{0}\vec{C}\right\}\right|=\boxed{[\vec{A}\vec{B}\vec{C}]^2}$$

- **Proof 3 (Transversal Cutting Two Forces, 2010–11 Q5a):** *Forces $\vec{P},\vec{Q}$ at O have resultant $\vec{R}=\vec{P}+\vec{Q}$, and a transversal line cuts their lines of action at A,B,C. Prove $\frac{|\vec{P}|}{OA}+\frac{|\vec{Q}|}{OB}=\frac{|\vec{R}|}{OC}$.*
  Proof: Let $\hat{u},\hat{v},\hat{w}$ be unit vectors along $\vec{P},\vec{Q},\vec{R}$. Then $\vec{OA}=OA\,\hat{u}=\frac{OA}{|\vec{P}|}\vec{P}$, $\vec{OB}=\frac{OB}{|\vec{Q}|}\vec{Q}$, and $\vec{OC}=\frac{OC}{|\vec{R}|}\vec{R}$. Since $\vec{P}+\vec{Q}=\vec{R}$:

$$\frac{|\vec{P}|}{OA}\vec{OA}+\frac{|\vec{Q}|}{OB}\vec{OB}=\frac{|\vec{R}|}{OC}\vec{OC}\implies\frac{|\vec{P}|}{OA}\vec{OA}+\frac{|\vec{Q}|}{OB}\vec{OB}-\frac{|\vec{R}|}{OC}\vec{OC}=\vec{0}$$

  Since A,B,C are collinear on the transversal line not passing through O, the sum of the coefficients in this linear relation among $\vec{OA},\vec{OB},\vec{OC}$ must be zero: $\boxed{\frac{|\vec{P}|}{OA}+\frac{|\vec{Q}|}{OB}=\frac{|\vec{R}|}{OC}}$.

- **Standard Vector Equations Solved for $\vec{r}$ or $\vec{x}$:**

  1. $\vec{r}\times\vec{b}=\vec{a}\times\vec{b}$ (2012–13 Q5c): $(\vec{r}-\vec{a})\times\vec{b}=\vec{0}\implies\vec{r}-\vec{a}\parallel\vec{b}\implies\boxed{\vec{r}=\vec{a}+t\vec{b}}$ ($t\in\mathbb{R}$).
  2. $k\vec{r}+\vec{a}\times\vec{r}=\vec{b}$ ($k\neq0$, 2014–15 Q5c, 2020–21 Q6b):
     - Dotting with $\vec{a}$: $k(\vec{a}\cdot\vec{r})+0=\vec{a}\cdot\vec{b}\implies\vec{a}\cdot\vec{r}=\frac{\vec{a}\cdot\vec{b}}{k}$.
     - Crossing with $\vec{a}$: $k(\vec{a}\times\vec{r})+\vec{a}\times(\vec{a}\times\vec{r})=\vec{a}\times\vec{b}\implies k(\vec{b}-k\vec{r})+(\vec{a}\cdot\vec{r})\vec{a}-a^2\vec{r}=\vec{a}\times\vec{b}$.
     - Substituting $\vec{a}\cdot\vec{r}=\frac{\vec{a}\cdot\vec{b}}{k}$ and solving for $\vec{r}$:

$$\boxed{\vec{r}=\frac{k^2\vec{b}+(\vec{a}\cdot\vec{b})\vec{a}-k(\vec{a}\times\vec{b})}{k(k^2+a^2)}}\quad\left(\text{Note: }+k(\vec{a}\times\vec{b})\text{ if }k\vec{r}+\vec{r}\times\vec{a}=\vec{b}\right)\qquad(24)$$

  3. $\vec{a}\times\vec{x}+(\vec{a}\cdot\vec{x})\vec{a}+\vec{b}=\vec{0}$ (2013–14 Q5c, 2018–19 Q6c):
     - Dotting with $\vec{a}$: $0+(\vec{a}\cdot\vec{x})a^2+\vec{a}\cdot\vec{b}=0\implies\vec{a}\cdot\vec{x}=-\frac{\vec{a}\cdot\vec{b}}{a^2}$.
     - Crossing $\vec{a}\times\vec{x}=-\vec{b}-(\vec{a}\cdot\vec{x})\vec{a}$ from the left with $\vec{a}$ gives $(\vec{a}\cdot\vec{x})\vec{a}-a^2\vec{x}=-\vec{a}\times\vec{b}$, yielding:

$$\boxed{\vec{x}=\frac{\vec{a}\times\vec{b}}{a^2}-\frac{(\vec{a}\cdot\vec{b})\vec{a}}{a^4}}\qquad(25)$$

5 Differential Geometry of Space Curves (Frenet-Serret Apparatus)

5.1 Space Curves, Curvature, Torsion & Frenet-Serret Formulae

Exam Appearances: `2010-11 Q6a`, `2011-12 Q6a,b`, `2012-13 Q6c`, `2013-14 Q6a`, `2014-15 Q6a (10)`, `2016-17 Q6a (10)`, `2017-18 Q6a,b`, `2018-19 Q6a`, `2019-20 Q5b`, `2020-21 Q6a`, `2021-22 Q6a`, `2022-23 Q6a`, `2023-24 Q6a,b`, `2024-25 Q6b`.
Textbook Citations: M. R. Spiegel, *Schaum's Outline of Vector Analysis*, Ch. 3, Sec. 3.4 (Geometry and Mechanics).

1. Definitions and Master Formulas for Space Curves

- **Space Curve (`2010-11 Q6a`):** If the position vector $\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$ of a moving point $P(x, y, z)$ in three-dimensional space is a continuous function of a single scalar parameter t, the locus traced by the terminal point of $\vec{r}(t)$ is called a **space curve** (or twisted curve if it does not lie in a single plane).
- **Arc Length Parameter (s):** $ds = |d\vec{r}| = \sqrt{(dx)^2 + (dy)^2 + (dz)^2} = |\vec{r}'(t)|\, dt \implies \frac{ds}{dt} = |\vec{r}'(t)|$.
- **Unit Tangent Vector ($\vec{T}$):** The unit vector in the direction of the tangent to the space curve:

$$\boxed{\vec{T} = \frac{d\vec{r}}{ds} = \frac{\vec{r}'(t)}{|\vec{r}'(t)|}} \tag{26}$$

- **Curvature (κ) and Radius of Curvature ($\rho = \frac{1}{\kappa}$):** The magnitude of the rate of change of the unit tangent vector $\vec{T}$ with respect to arc length s (measuring how rapidly the curve bends away from its tangent line):

$$\boxed{\kappa = \left|\frac{d\vec{T}}{ds}\right| = \frac{|\vec{T}'(t)|}{|\vec{r}'(t)|} = \frac{|\vec{r}'(t) \times \vec{r}''(t)|}{|\vec{r}'(t)|^3}}, \qquad \rho = \frac{1}{\kappa} \tag{27}$$

- **Unit Principal Normal Vector ($\vec{N}$):** The unit vector in the direction of $\frac{d\vec{T}}{ds}$ (perpendicular to $\vec{T}$ and pointing toward the center of curvature):

$$\boxed{\vec{N} = \frac{1}{\kappa}\frac{d\vec{T}}{ds} = \frac{\vec{T}'(t)}{|\vec{T}'(t)|} = \vec{B} \times \vec{T}} \tag{28}$$

- **Unit Binormal Vector ($\vec{B}$):** The unit vector perpendicular to both $\vec{T}$ and $\vec{N}$ such that $(\vec{T}, \vec{N}, \vec{B})$ forms a right-handed orthonormal triad:

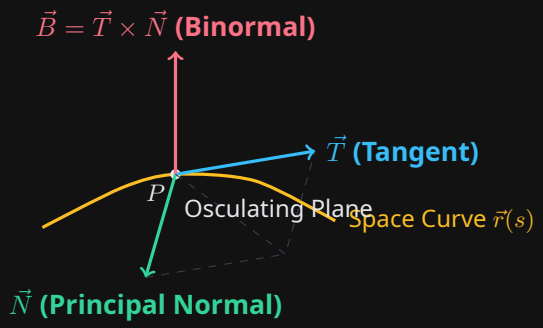
$$\boxed{\vec{B} = \vec{T} \times \vec{N} = \frac{\vec{r}'(t) \times \vec{r}''(t)}{|\vec{r}'(t) \times \vec{r}''(t)|}} \tag{29}$$

- **Torsion (τ) and Radius of Torsion ($\sigma = \frac{1}{\tau}$):** The rate at which the osculating plane twists out of its plane with respect to arc length s (defined by $\frac{d\vec{B}}{ds} = -\tau\vec{N}$):

$$\boxed{\tau = -\vec{N} \cdot \frac{d\vec{B}}{ds} = \frac{[\vec{r}'(t)\ \ \vec{r}''(t)\ \ \vec{r}'''(t)]}{|\vec{r}'(t) \times \vec{r}''(t)|^2}}, \qquad \sigma = \frac{1}{\tau} \tag{30}$$

- **Equations of Fundamental Lines and Planes at Point $\vec{r}_0$ ($\vec{R} = x\hat{i} + y\hat{j} + z\hat{k}$):**
  - *Tangent Line:* $\vec{R} = \vec{r}_0 + \lambda\vec{T}$; *Principal Normal Line:* $\vec{R} = \vec{r}_0 + \lambda\vec{N}$; *Binormal Line:* $\vec{R} = \vec{r}_0 + \lambda\vec{B}$.
  - *Osculating Plane* (contains $\vec{T}, \vec{N}$; normal $\vec{B}$): $(\vec{R} - \vec{r}_0) \cdot \vec{B} = 0$.

- *Normal Plane* (contains $\vec{N}, \vec{B}$; normal $\vec{T}$): $(\vec{R} - \vec{r}_0) \cdot \vec{T} = 0$.
- *Rectifying Plane* (contains $\vec{T}, \vec{B}$; normal $\vec{N}$): $(\vec{R} - \vec{r}_0) \cdot \vec{N} = 0$.



2. Derivation of the Frenet-Serret Formulae (`2014-15 Q6a`, `2016-17 Q6a`, `2023-24 Q6a`)

The vectors $(\vec{T}, \vec{N}, \vec{B})$ form a right-handed mutually orthogonal triad of unit vectors:

$$\vec{T} \cdot \vec{T} = \vec{N} \cdot \vec{N} = \vec{B} \cdot \vec{B} = 1, \qquad \vec{T} \cdot \vec{N} = \vec{N} \cdot \vec{B} = \vec{B} \cdot \vec{T} = 0$$

1. **First Formula ($\frac{d\vec{T}}{ds} = \kappa\vec{N}$):** Differentiating $\vec{T} \cdot \vec{T} = 1$ with respect to s gives $2\vec{T} \cdot \frac{d\vec{T}}{ds} = 0 \implies \frac{d\vec{T}}{ds} \perp \vec{T}$. By definition, the unit vector along $\frac{d\vec{T}}{ds}$ is the principal normal $\vec{N}$ and its magnitude is the curvature κ:

$$\boxed{\frac{d\vec{T}}{ds} = \kappa\vec{N}} \tag{31}$$

2. **Third Formula ($\frac{d\vec{B}}{ds} = -\tau\vec{N}$):**

   - Differentiating $\vec{B} \cdot \vec{B} = 1$ gives $2\vec{B} \cdot \frac{d\vec{B}}{ds} = 0 \implies \frac{d\vec{B}}{ds} \perp \vec{B}$.
   - Differentiating $\vec{B} = \vec{T} \times \vec{N}$ with respect to s:

$$\frac{d\vec{B}}{ds} = \frac{d\vec{T}}{ds} \times \vec{N} + \vec{T} \times \frac{d\vec{N}}{ds} = (\kappa\vec{N}) \times \vec{N} + \vec{T} \times \frac{d\vec{N}}{ds} = \vec{T} \times \frac{d\vec{N}}{ds}$$

   Thus $\frac{d\vec{B}}{ds} \perp \vec{T}$ as well. Since $\frac{d\vec{B}}{ds}$ is perpendicular to both $\vec{T}$ and $\vec{B}$, it must be parallel to $\vec{N}$:

$$\boxed{\frac{d\vec{B}}{ds} = -\tau\vec{N}} \tag{32}$$

   where the proportionality scalar τ is defined as the **torsion**.

3. **Second Formula ($\frac{d\vec{N}}{ds} = \tau\vec{B} - \kappa\vec{T}$):** Since $\vec{N} = \vec{B} \times \vec{T}$, differentiate with respect to s:

$$\begin{aligned} \frac{d\vec{N}}{ds} &= \frac{d\vec{B}}{ds} \times \vec{T} + \vec{B} \times \frac{d\vec{T}}{ds} = (-\tau\vec{N}) \times \vec{T} + \vec{B} \times (\kappa\vec{N}) \\ &= -\tau(-\vec{B}) + \kappa(-\vec{T}) \implies \boxed{\frac{d\vec{N}}{ds} = \tau\vec{B} - \kappa\vec{T}} \end{aligned} \tag{33}$$

3. Circular Motion Kinematics Proof (`2011-12 Q6a`, `2017-18 Q6a`)

Given $\vec{r}(t) = \cos\omega t\,\hat{i} + \sin\omega t\,\hat{j}$ (ω = constant), where $r = |\vec{r}| = \sqrt{\cos^2\omega t + \sin^2\omega t} = 1$:

1. **(i) Velocity $\vec{v} \perp \vec{r}$:** $\vec{v} = \frac{d\vec{r}}{dt} = -\omega\sin\omega t\,\hat{i} + \omega\cos\omega t\,\hat{j}$. Taking the dot product:

$$\vec{r}\cdot\vec{v} = -\omega\cos\omega t\sin\omega t + \omega\sin\omega t\cos\omega t = 0 \implies \boxed{\vec{v}\perp\vec{r}}$$

2. **(ii) Acceleration $\vec{a}$ Directed Toward Origin:** $\vec{a} = \frac{d\vec{v}}{dt} = -\omega^2\cos\omega t\,\hat{i} - \omega^2\sin\omega t\,\hat{j} = \boxed{-\omega^2\vec{r}}$. The negative sign shows $\vec{a}$ is directed opposite to $\vec{r}$ (toward the origin) with magnitude $|\vec{a}| = \omega^2|\vec{r}| \propto |\vec{r}|$.
3. **(iii) $\vec{r}\times\vec{v}$ is Constant:**

$$\vec{r}\times\vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos\omega t & \sin\omega t & 0 \\ -\omega\sin\omega t & \omega\cos\omega t & 0 \end{vmatrix} = \omega(\cos^2\omega t + \sin^2\omega t)\hat{k} = \boxed{\omega\hat{k} = \text{constant vector}}$$

6 Gradient, Divergence, Curl, Potentials & Vector Differential Identities

6.1 Directional/Normal Derivatives, Physical Interpretations & Vector Identities

Exam Appearances: 2010-11 Q6c, Q7a, 2011-12 Q7a, 2012-13 Q6b, Q7a, 2013-14 Q6b, Q7a, 2014-15 Q6b,c, 2016-17 Q6b,c, 2017-18 Q5c, Q6c, Q7a, 2018-19 Q5c, Q7a,b,c, 2019-20 Q6a,b, 2020-21 Q6c, Q7a, 2021-22 Q6b,c, 2022-23 Q6b,c, 2023-24 Q6c, 2024-25 Q7a.
Textbook Citations: M. R. Spiegel, *Schaum's Outline of Vector Analysis*, Ch. 4 (Gradient, Divergence and Curl).

1. Definitions: Directional, Normal & Tangential Derivatives

- **Directional Derivative (**2011-12 Q7a, 2014-15 Q6b, 2016-17 Q6b, 2020-21 Q6c**):** The rate of change of a differentiable scalar field $\phi(x, y, z)$ with respect to distance s at a point $P(x, y, z)$ in the direction of a given unit vector $\hat{a} = \frac{\vec{a}}{|\vec{a}|}$ is called the **directional derivative** of ϕ in the direction of $\hat{a}$:

$$\boxed{D_{\hat{a}}\phi = \frac{d\phi}{ds} = \nabla\phi \cdot \hat{a} = |\nabla\phi| \cos\theta} \tag{34}$$

  where $\nabla\phi = \frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j} + \frac{\partial\phi}{\partial z}\hat{k}$.
  - *Maximum Rate of Increase:* $D_{\hat{a}}\phi$ is greatest when $\cos\theta = +1$ ($\theta = 0$), i.e., in the **direction of** $\nabla\phi$, and its maximum value is $\boxed{|\nabla\phi|}$.
  - *Most Rapid Decrease (2019-20 Q6b):* Temperature/potential drops most rapidly in the direction of $\boxed{-\nabla\phi}$, with rate of change $-|\nabla\phi|$.
- **Normal Derivative (**2011-12 Q7a, 2014-15 Q6b, 2018-19 Q7c**):** The directional derivative of ϕ taken along the unit normal vector $\hat{n}$ to a level surface $\phi(x, y, z) = c$: $\boxed{\frac{d\phi}{dn} = \nabla\phi \cdot \hat{n} = |\nabla\phi|}$.
- **Tangential Derivative (**2018-19 Q7c**):** The directional derivative of ϕ taken along the unit tangent vector $\vec{T} = \frac{d\vec{r}}{ds}$ to a curve C: $\boxed{\frac{d\phi}{ds} = \nabla\phi \cdot \vec{T}}$. (Along a level surface $\phi = c$, $\vec{T} \perp \nabla\phi \implies \frac{d\phi}{ds} = 0$).

2. Physical Interpretation of Divergence and Curl, Solenoidal & Conservative Fields (2020-21 Q7a, 2022-23 Q6b, 2024-25 Q7a)

- **Physical Interpretation of Divergence (**$\text{div}\,\vec{F} = \nabla \cdot \vec{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$**):** $\nabla \cdot \vec{F}$ at a point P measures the **net outward flux of the vector field** $\vec{F}$ **per unit volume** emerging from an infinitesimal volume element ΔV enclosing P: $\nabla \cdot \vec{F} = \lim_{\Delta V \to 0} \frac{1}{\Delta V} \iint_{\Delta S} \vec{F} \cdot \hat{n}\, dS$.
  - If $\nabla \cdot \vec{F} > 0$, P is a **source** (fluid/flux is expanding or being created, e.g., positive electric charge $\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$ or expanding heated gas).
  - If $\nabla \cdot \vec{F} < 0$, P is a **sink** (fluid is compressing or being absorbed).
  - **Solenoidal Vector Field:** If $\boxed{\nabla \cdot \vec{F} = 0}$ everywhere, the field has no sources or sinks, and flux entering any closed surface equals flux leaving it (e.g., incompressible fluid velocity $\vec{v}$, or magnetic field $\nabla \cdot \vec{B} = 0$). A solenoidal field can be derived from a **vector potential** $\vec{V}$ such that $\vec{F} = \nabla \times \vec{V}$.
- **Physical Interpretation of Curl (**$\text{curl}\,\vec{F} = \nabla \times \vec{F}$**):** $\nabla \times \vec{F}$ measures the **microscopic circulation (vorticity or local rotational tendency)** of the vector field $\vec{F}$ per unit area around a point P. For a rigid body rotating with angular velocity $\vec{\omega}$, the linear velocity is $\vec{v} = \vec{\omega} \times \vec{r}$ and

$\nabla \times \vec{v} = 2\vec{\omega}$ (twice the angular velocity vector). A paddle wheel placed in a fluid flow $\vec{v}$ rotates if $\nabla \times \vec{v} \neq \vec{0}$.

- **Irrotational / Conservative Force Field (2024–25 Q7a):** A vector field $\vec{F}$ is called **irrotational** (or a **conservative force field**) in a simply connected region if its curl vanishes identically:

$$\boxed{\nabla \times \vec{F} = \vec{0} \iff \vec{F} = \nabla\phi \quad (\text{or } -\nabla\phi)} \tag{35}$$

where $\phi(x, y, z)$ is the single-valued **scalar potential**. In a conservative force field, the work done $\int_A^B \vec{F} \cdot d\vec{r} = \phi(B) - \phi(A)$ is **independent of the path** joining A and B, and $\oint_C \vec{F} \cdot d\vec{r} = 0$ around any closed loop C.

3. Complete Proofs of All Tested Vector Differential Identities

For position vector $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $r = |\vec{r}| = \sqrt{x^2 + y^2 + z^2}$, recall the fundamental building blocks:

$$\frac{\partial r}{\partial x} = \frac{x}{r}, \qquad \nabla r = \frac{\vec{r}}{r}, \qquad \nabla f(r) = f'(r)\nabla r = \frac{f'(r)}{r}\vec{r}, \qquad \nabla \cdot \vec{r} = 3, \qquad \nabla \times \vec{r} = \vec{0}$$

1. **Identity 1 (2010–11 Q6c, 2013–14 Q6b):** *Prove* $\text{div}\left\{\frac{f(r)\vec{r}}{r}\right\} = \frac{1}{r^2}\frac{d}{dr}\left\{r^2 f(r)\right\}$.
   Proof: Using the product rule $\nabla \cdot (\phi\vec{A}) = (\nabla\phi) \cdot \vec{A} + \phi(\nabla \cdot \vec{A})$ with $\phi = \frac{f(r)}{r}$ and $\vec{A} = \vec{r}$:

$$\begin{aligned}
\nabla \cdot \left(\frac{f(r)}{r}\vec{r}\right) &= \left[\nabla\left(\frac{f(r)}{r}\right)\right] \cdot \vec{r} + \frac{f(r)}{r}(\nabla \cdot \vec{r}) \\
&= \left[\frac{rf'(r) - f(r)}{r^2}\frac{\vec{r}}{r}\right] \cdot \vec{r} + 3\frac{f(r)}{r} \qquad (\text{since } \vec{r} \cdot \vec{r} = r^2) \\
&= f'(r) - \frac{f(r)}{r} + 3\frac{f(r)}{r} = f'(r) + \frac{2}{r}f(r) = \frac{1}{r^2}\left[r^2 f'(r) + 2rf(r)\right] = \boxed{\frac{1}{r^2}\frac{d}{dr}\left\{r^2 f(r)\right\}}
\end{aligned}$$

2. **Identity 2 (2017–18 Q5c, 2021–22 Q6b):** *Prove* $\nabla \cdot \left(\frac{\vec{r}}{r^3}\right) = 0$ *(for* $r \neq 0$*) and find the most general* $f(r)$ *such that* $f(r)\vec{r}$ *is solenoidal.*
   Proof: Using $\nabla \cdot (f(r)\vec{r}) = \left(\frac{f'(r)}{r}\vec{r}\right) \cdot \vec{r} + f(r)(\nabla \cdot \vec{r}) = rf'(r) + 3f(r)$:
   - For $f(r) = r^{-3}$: $\nabla \cdot \left(r^{-3}\vec{r}\right) = r(-3r^{-4}) + 3r^{-3} = -3r^{-3} + 3r^{-3} = \boxed{0}$.
   - For $f(r)\vec{r}$ to be solenoidal: $rf'(r) + 3f(r) = 0 \implies \frac{df}{f} = -3\frac{dr}{r} \implies \ln|f| = -3\ln r + \ln C \implies \boxed{f(r) = \frac{C}{r^3}}$.

3. **Identity 3 (2016–17 Q6c):** *Prove* $\nabla^2 f(r) = f''(r) + \frac{2}{r}f'(r)$.
   Proof: Since $\nabla^2 f(r) = \nabla \cdot (\nabla f(r)) = \nabla \cdot \left(\frac{f'(r)}{r}\vec{r}\right)$, apply Identity 1 (replacing $f(r)$ by $f'(r)$):

$$\nabla^2 f(r) = \nabla \cdot \left(\frac{f'(r)}{r}\vec{r}\right) = \frac{1}{r^2}\frac{d}{dr}\left[r^2 f'(r)\right] = \frac{1}{r^2}\left[r^2 f''(r) + 2rf'(r)\right] = \boxed{f''(r) + \frac{2}{r}f'(r)}$$

4. **Identity 4 (2018–19 Q7a):** *Show that* $\nabla^2(r^n\vec{r}) = n(n+3)r^{n-2}\vec{r}$.
   Proof: Take the x-component $r^n x$. Differentiating partially with respect to x:

$$\begin{aligned}
\frac{\partial}{\partial x}(r^n x) &= r^n + nr^{n-1}\left(\frac{x}{r}\right)x = r^n + nr^{n-2}x^2 \\
\frac{\partial^2}{\partial x^2}(r^n x) &= nr^{n-2}x + n(n-2)r^{n-4}x^3 + 2nr^{n-2}x = 3nr^{n-2}x + n(n-2)r^{n-4}x^3
\end{aligned}$$

Similarly, for y and z:

$$\frac{\partial^2}{\partial y^2}(r^n x) = nr^{n-2}x + n(n-2)r^{n-4}xy^2, \qquad \frac{\partial^2}{\partial z^2}(r^n x) = nr^{n-2}x + n(n-2)r^{n-4}xz^2$$

Summing all three second derivatives (and using $x^2+y^2+z^2=r^2$):

$$\nabla^2(r^n x) = 5nr^{n-2}x + n(n-2)r^{n-4}x(r^2) = n(n+3)r^{n-2}x$$

By symmetry in x, y, z, combining all three components gives $\boxed{\nabla^2(r^n\vec{r}) = n(n+3)r^{n-2}\vec{r}}$.

5. **Identity 5 (**`2012-13 Q6b`**,** `2014-15 Q6c`**):** *Prove* $\nabla\times(\nabla\times\vec{A}) = \nabla(\nabla\cdot\vec{A}) - \nabla^2\vec{A}$.
Proof: Let $\vec{A} = A_1\hat{i} + A_2\hat{j} + A_3\hat{k}$. Then $\nabla\times\vec{A} = \left(\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z}\right)\hat{i} + \left(\frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x}\right)\hat{j} + \left(\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y}\right)\hat{k}$.
Taking the x-component of $\nabla\times(\nabla\times\vec{A})$:

$$\begin{aligned}[\nabla\times(\nabla\times\vec{A})]_x &= \frac{\partial}{\partial y}\left(\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y}\right) - \frac{\partial}{\partial z}\left(\frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x}\right)\\ &= \left(\frac{\partial^2 A_1}{\partial x^2} + \frac{\partial^2 A_2}{\partial x\partial y} + \frac{\partial^2 A_3}{\partial x\partial z}\right) - \left(\frac{\partial^2 A_1}{\partial x^2} + \frac{\partial^2 A_1}{\partial y^2} + \frac{\partial^2 A_1}{\partial z^2}\right)\\ &= \frac{\partial}{\partial x}(\nabla\cdot\vec{A}) - \nabla^2 A_1\end{aligned}$$

By cyclic symmetry for the y- and z-components, $\boxed{\nabla\times(\nabla\times\vec{A}) = \nabla(\nabla\cdot\vec{A}) - \nabla^2\vec{A}}$.

6. **Identity 6 (**`2018-19 Q5c` **&** `2020-21 Q7a`**):** *Prove* $\nabla\cdot(\vec{A}\times\vec{B}) = \vec{B}\cdot(\nabla\times\vec{A}) - \vec{A}\cdot(\nabla\times\vec{B})$*, and evaluate* $\nabla\cdot(\vec{A}\times\vec{r})$ *if* $\nabla\times\vec{A} = \vec{0}$.
Proof: Expanding in components:

$$\begin{aligned}\nabla\cdot(\vec{A}\times\vec{B}) &= \frac{\partial}{\partial x}(A_2B_3 - A_3B_2) + \frac{\partial}{\partial y}(A_3B_1 - A_1B_3) + \frac{\partial}{\partial z}(A_1B_2 - A_2B_1)\\ &= B_1\left(\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z}\right) + B_2\left(\frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x}\right) + B_3\left(\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y}\right)\\ &\quad - A_1\left(\frac{\partial B_3}{\partial y} - \frac{\partial B_2}{\partial z}\right) - A_2\left(\frac{\partial B_1}{\partial z} - \frac{\partial B_3}{\partial x}\right) - A_3\left(\frac{\partial B_2}{\partial x} - \frac{\partial B_1}{\partial y}\right)\\ &= \boxed{\vec{B}\cdot(\nabla\times\vec{A}) - \vec{A}\cdot(\nabla\times\vec{B})}\end{aligned}$$

Corollary (`2020-21 Q7a`*):* Setting $\vec{B} = \vec{r}$ (where $\nabla\times\vec{r} = \vec{0}$) and given $\nabla\times\vec{A} = \vec{0}$:

$$\nabla\cdot(\vec{A}\times\vec{r}) = \vec{r}\cdot(\nabla\times\vec{A}) - \vec{A}\cdot(\nabla\times\vec{r}) = \vec{r}\cdot\vec{0} - \vec{A}\cdot\vec{0} = \boxed{0}$$

7. **Identity 7 (**`2017-18 Q6c`**):** *Differentiate* $\frac{\vec{r}}{\vec{r}\cdot\vec{a}} - \frac{\vec{r}\times\vec{a}}{r}$ *with respect to* t *(*$\vec{a} = $ *const*$, r = |\vec{r}|$*,* $\frac{dr}{dt} = \frac{\vec{r}\cdot\dot{\vec{r}}}{r}$*):*

$$\frac{d}{dt}\left(\frac{\vec{r}}{\vec{r}\cdot\vec{a}} - \frac{\vec{r}\times\vec{a}}{r}\right) = \boxed{\frac{(\vec{r}\cdot\vec{a})\dot{\vec{r}} - (\dot{\vec{r}}\cdot\vec{a})\vec{r}}{(\vec{r}\cdot\vec{a})^2} - \frac{r^2(\dot{\vec{r}}\times\vec{a}) - (\vec{r}\cdot\dot{\vec{r}})(\vec{r}\times\vec{a})}{r^3}}$$

7 Integral Theorems: Line Integrals, Green's, Stokes' & Gauss's Theorems

7.1 Statements, Deductions, Proofs & Applications of the Integral Theorems

Exam Appearances: 2010-11 Q7c, Q8a,b, 2011-12 Q8b, 2012-13 Q8b, 2013-14 Q7b, 2014-15 Q8a, 2016-17 Q8a, 2017-18 Q8a, 2018-19 Q8a, 2020-21 Q8a, 2021-22 Q8a,b, 2022-23 Q8a, 2023-24 Q7a, Q8a, 2024-25 Q7c, Q8a,b.
Textbook Citations: M. R. Spiegel, *Schaum's Outline of Vector Analysis*, Ch. 5 & Ch. 6; H. Anton, I. Bivens & S. Davis, *Calculus* (10th Ed.), Ch. 15.

1. Definition of Line Integral of a Vector-Valued Function (2023-24 Q7a)

Let $\vec{F}(x,y,z) = F_1(x,y,z)\hat{i} + F_2(x,y,z)\hat{j} + F_3(x,y,z)\hat{k}$ be a continuous vector field defined along a smooth curve C given by $\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$ ($a \le t \le b$). Partitioning C into n sub-arcs $\Delta\vec{r}_k$ and taking the limit of the Riemann sum of tangential components gives the **line integral of $\vec{F}$ along C**:

$$\boxed{\int_C \vec{F}\cdot d\vec{r} = \int_C (F_1\,dx + F_2\,dy + F_3\,dz) = \int_a^b \vec{F}(\vec{r}(t))\cdot\vec{r}\,'(t)\,dt = \int_C (\vec{F}\cdot\vec{T})\,ds} \tag{36}$$

Physically, if $\vec{F}$ represents a force field, $\int_C \vec{F}\cdot d\vec{r}$ represents the **total work done** by the force in moving a particle along C.

2. Green's Theorem in the Plane, Applications & Deduction from Stokes' Theorem (2011-12 Q8b, 2013-14 Q7b, 2021-22 Q8a, 2022-23 Q8a, 2023-24 Q8a, 2024-25 Q7c)

Statement of Green's Theorem in the Plane: Let R be a closed bounded region in the xy-plane whose boundary C consists of one or more piecewise smooth, simple closed curves oriented positively (counterclockwise). If $M(x,y)$ and $N(x,y)$ are continuous and have continuous first partial derivatives $\frac{\partial M}{\partial y}$ and $\frac{\partial N}{\partial x}$ throughout R, then:

$$\boxed{\oint_C (M\,dx + N\,dy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy} \tag{37}$$

Deduction of Green's Theorem from Stokes' Theorem (2011-12 Q8b, 2013-14 Q7b):

1. By Stokes' theorem for a surface S bounded by C:
$$\oint_C \vec{F}\cdot d\vec{r} = \iint_S (\nabla\times\vec{F})\cdot\hat{n}\,dS$$
2. Let S be a flat planar region R lying entirely in the xy-plane ($z = 0, dz = 0$). Then the upward unit normal is $\hat{n} = \hat{k}$, the surface area element is $dS = dx\,dy$, and $d\vec{r} = dx\,\hat{i} + dy\,\hat{j}$.
3. Choose the 2D vector field $\vec{F}(x,y) = M(x,y)\hat{i} + N(x,y)\hat{j} + 0\hat{k}$. Then:
$$\vec{F}\cdot d\vec{r} = M\,dx + N\,dy, \qquad \text{and} \qquad \nabla\times\vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ M(x,y) & N(x,y) & 0 \end{vmatrix} = \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right)\hat{k}$$
4. Taking the dot product with $\hat{n} = \hat{k}$ gives $(\nabla\times\vec{F})\cdot\hat{n} = \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}$. Substituting into Stokes' theorem immediately yields Green's theorem in the plane:
$$\oint_C (M\,dx + N\,dy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy$$

Applications of Green's Theorem in the Plane (2021–22 Q8a, 2022–23 Q8a, 2023–24 Q8a):

1. **Computing the Area of a Plane Region Bounded by a Closed Curve C:** Choosing $(M, N) = (0, x)$, $(-y, 0)$, or $(-\frac{1}{2}y, \frac{1}{2}x)$ gives $\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = 1$, so the area of R is:

$$\boxed{\text{Area}(R) = \iint_R dx\,dy = \oint_C x\,dy = -\oint_C y\,dx = \frac{1}{2}\oint_C (x\,dy - y\,dx)} \tag{38}$$

2. **Evaluating Difficult Line Integrals / Work Done:** Converts complicated closed-loop line integrals $\oint_C \vec{F} \cdot d\vec{r}$ into simpler double integrals over the enclosed region R.
3. **Testing Path Independence & Proving Conservation Laws:** Shows that in a simply connected domain where $\frac{\partial N}{\partial x} = \frac{\partial M}{\partial y}$, $\oint_C (M\,dx + N\,dy) = 0$ (exact differential / conservative 2D field), and proves Cauchy's integral theorem in complex analysis.

3. Statement of Stokes' Theorem (2010–11 Q8a, 2012–13 Q8b, 2017–18 Q8a, 2024–25 Q8b)

Statement: Let S be an open, two-sided piecewise smooth surface bounded by a simple closed curve C with positive orientation (right-hand rule relative to the unit normal $\hat{n}$ of S). If $\vec{F}(x, y, z)$ is a vector field having continuous first-order partial derivatives on S and C, then the circulation of $\vec{F}$ around C equals the flux of $\nabla \times \vec{F}$ across S:

$$\boxed{\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{n}\,dS} \tag{39}$$

4. Statement, Physical Interpretation & Proof of Gauss's Divergence Theorem (2010–11 Q8b, 2014–15 Q8a, 2020–21 Q8a, 2021–22 Q8b, 2024–25 Q8a)

Statement: Let V be a closed, bounded three-dimensional solid region enclosed by a piecewise smooth, orientable closed surface S with outward-pointing unit normal $\hat{n}$. If $\vec{F}(x, y, z) = F_1\hat{i} + F_2\hat{j} + F_3\hat{k}$ is a vector field continuous with continuous first-order partial derivatives throughout V and on S, then:

$$\boxed{\iint_S \vec{F} \cdot \hat{n}\,dS = \iiint_V (\nabla \cdot \vec{F})\,dV} \tag{40}$$

Physical Interpretation (2020–21 Q8a): For a fluid flow or flux field $\vec{F}$, Gauss's divergence theorem states that the **total outward flux** of $\vec{F}$ escaping across the closed boundary surface S per unit time is equal to the **volume integral of all net sources and sinks ($\nabla \cdot \vec{F}$)** distributed throughout the interior volume V.

Proof of Gauss's Divergence Theorem (2010–11 Q8b): In component form, $\iint_S \vec{F}\cdot\hat{n}\,dS = \iint_S \left(F_1\hat{i}\cdot\hat{n} + F_2\hat{j}\cdot\hat{n}\right.$ and $\iiint_V (\nabla \cdot \vec{F})\,dV = \iiint_V \left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}\right) dV$. Thus, it suffices to prove:

$$\iint_S F_3(\hat{k} \cdot \hat{n})\,dS = \iiint_V \frac{\partial F_3}{\partial z}\,dx\,dy\,dz \tag{41}$$

(and the analogous relations for F_1 and F_2).

1. Let V be a simple solid region bounded below by a lower surface $S_1 : z = g_1(x, y)$ and above by an upper surface $S_2 : z = g_2(x, y)$ over a projection R in the xy-plane, with vertical cylindrical side walls S_3 (where $\hat{k} \cdot \hat{n} = 0$).
2. Evaluating the volume integral on the right-hand side of Equation (41) by integrating first with respect to z:

$$\iiint_V \frac{\partial F_3}{\partial z}\,dV = \iint_R \left[\int_{z=g_1(x,y)}^{z=g_2(x,y)} \frac{\partial F_3}{\partial z}\,dz\right] dx\,dy = \iint_R [F_3(x, y, g_2(x, y)) - F_3(x, y, g_1(x, y))]\,dx\,dy$$

3. On the upper surface S_2, the outward normal $\hat{n}_2$ points upward ($\cos\gamma_2 = \hat{k}\cdot\hat{n}_2 > 0$), so $\hat{k}\cdot\hat{n}_2\,dS = +dx\,dy$. On the lower surface S_1, the outward normal $\hat{n}_1$ points downward ($\cos\gamma_1 = \hat{k}\cdot\hat{n}_1 < 0$), so $\hat{k}\cdot\hat{n}_1\,dS = -dx\,dy$. On any vertical side wall S_3, $\hat{k}\cdot\hat{n}_3 = 0$.
4. Therefore:

$$\iint_S F_3(\hat{k}\cdot\hat{n})\,dS = \iint_{S_2} F_3(\hat{k}\cdot\hat{n}_2)\,dS + \iint_{S_1} F_3(\hat{k}\cdot\hat{n}_1)\,dS = \iint_R [F_3(x,y,g_2) - F_3(x,y,g_1)]\,dx\,dy = \iiint_V \frac{\partial F_3}{\partial z}\,dV$$

5. By exact symmetry along the x- and y-axes:

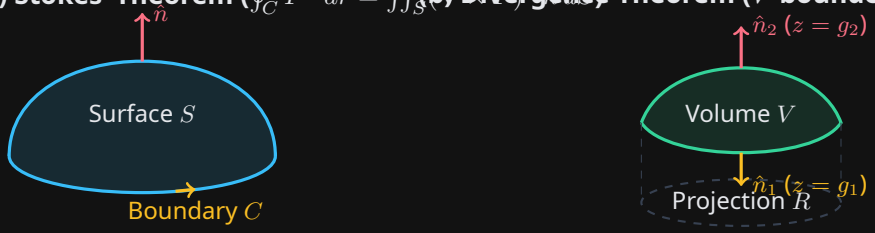
$$\iint_S F_1(\hat{\imath}\cdot\hat{n})\,dS = \iiint_V \frac{\partial F_1}{\partial x}\,dV \qquad \text{and} \qquad \iint_S F_2(\hat{\jmath}\cdot\hat{n})\,dS = \iiint_V \frac{\partial F_2}{\partial y}\,dV$$

Adding all three equations gives $\iint_S \vec{F}\cdot\hat{n}\,dS = \iiint_V (\nabla\cdot\vec{F})\,dV$.

Corollary (Poisson's Equation Flux Proof, 2010-11 Q7c): Given $\vec{F} = \nabla f$ and $\nabla^2 f = -4\pi\rho$, we have $\nabla\cdot\vec{F} = \nabla\cdot(\nabla f) = \nabla^2 f = -4\pi\rho$. Applying Gauss's Divergence Theorem immediately yields:

$$\iint_S \vec{F}\cdot\hat{n}\,dS = \iiint_V (\nabla\cdot\vec{F})\,dV = \iiint_V (\nabla^2 f)\,dV = \boxed{-4\pi \iiint_V \rho\,dV} \tag{42}$$

(a) Stokes' Theorem ($\oint_C \vec{F}\cdot d\vec{r} = \iint_S (\nabla\times\vec{F})\cdot\hat{n}\,dS$) **(b) Divergence Theorem (V bounded by $S = S_1 \cup S_2$)**



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4. **Anton, Howard, Bivens, Irl, & Davis, Stephen**, *Calculus: Early Transcendentals* (10th Edition), John Wiley & Sons.